CH510L

# SVM-based Classification of Bioelectrochemical Reactors

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# Problem Statement

## Challenge:

- Manual classification of Bio-reactor performance, based on input, is tedious.
- Need novel ways to classify the data, based on performance of reactor.

## Curiosity:

- Can ML techniques help in predicting the reactor type for suitable input (conc., pH, OD, etc.)

## Goal:

- Achieve high accuracy on the classification model, thus predicting the reactor.

# Technical Background

## Microbial Electrosynthesis (MES):

- A sustainable bioelectrochemical process that uses microorganisms to convert $CO_2$ into valuable chemicals by leveraging electrical energy.

## Key Parameters:

- Current (I, mA)
- Charge (Q, mC)
- Formate conc. (C1, mg/L)
- Acetate conc. (C2, mg/L)
- Ammonia conc. (NH4, mg/L)
- Optical Density (OD)
- pH

## Classification Criteria:

- Control Reactor (0%) - 0
- Low-Ni doped Reactor (0.01%) - 1
- High-Ni doped Reactor (5%) - 2

# Dataset Overview

- Collected from a controlled experiment.
- **Study:**
  - Effect of Ni doping on Bio-reactors in MES.
- The concentration of acetate and formate, on different days, and with different doping levels were observed.
- To further study the effects, electron recovery %, along with pH, NH4 conc. were observed.

**Samples**
**282**

**Reactors 6**

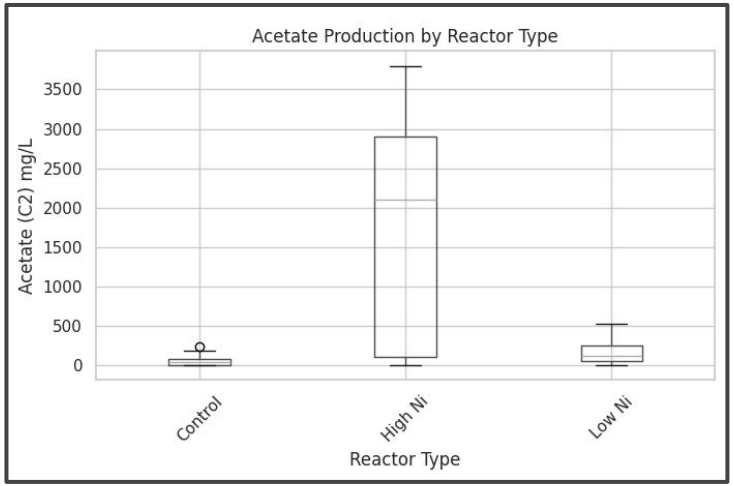
**Days**
**48**

**Features**
**14**

# EDA - Acetate Production

## Key Finding:

- Ni doping drastically increases Acetate production (C2), enhancing bioprocess synthesis.




| | C2_mg_L |
|---|---|
| count | 120.000000 |
| mean | 661.110120 |
| std | 1106.592425 |
| min | 0.000000 |
| 25% | 13.665128 |
| 50% | 97.305981 |
| 75% | 426.676280 |
| max | 3798.432526 |

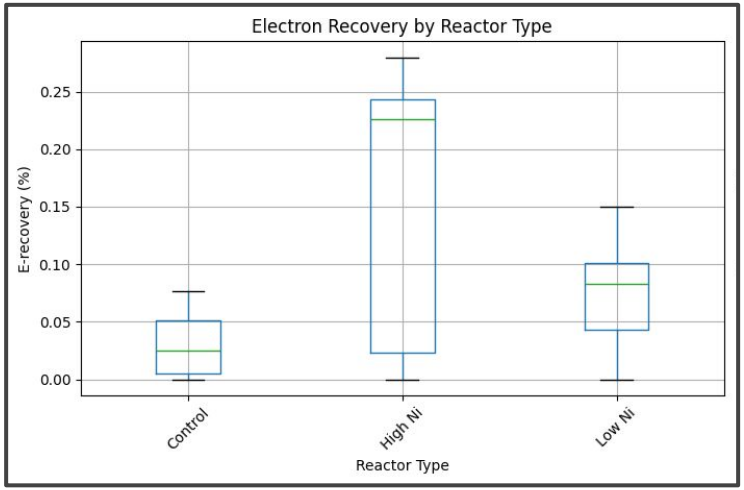
For each feature f:

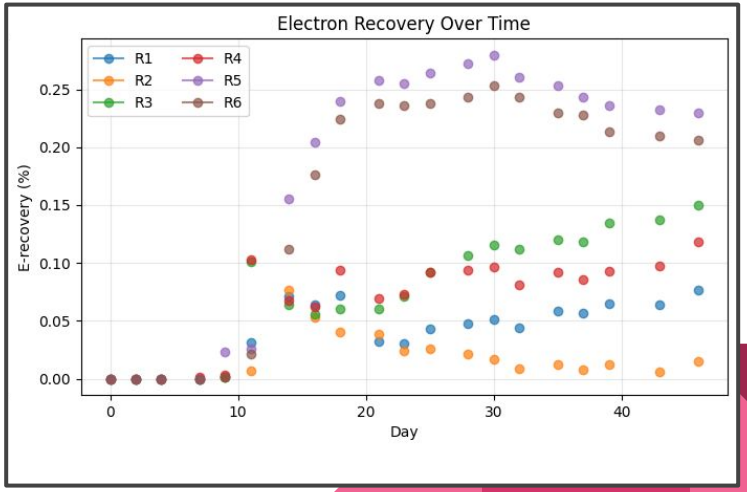
$$\mu_f = \frac{1}{n}\sum_{i=1}^{n} x_{if}, \quad \sigma_f = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_{if} - \mu_f)^2}$$

# EDA - Performance Metrics

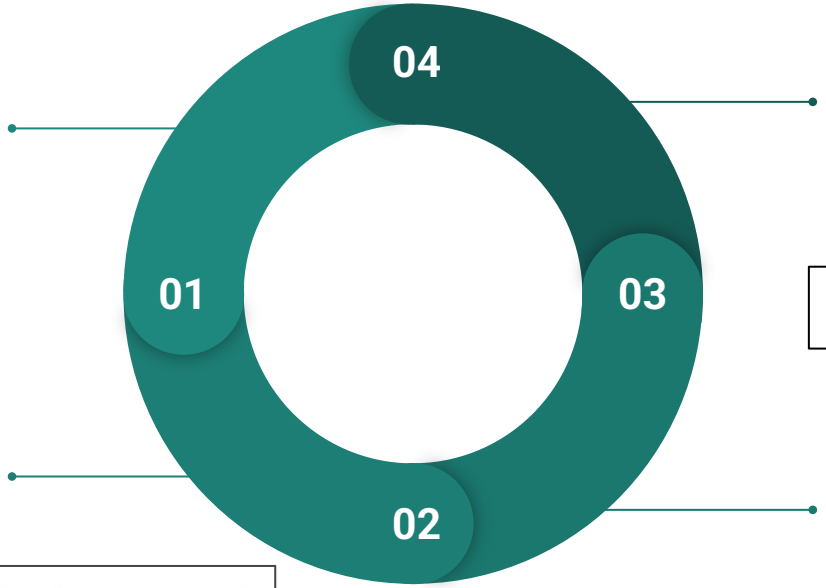
- High electron recovery shows strong Correlation between Ni doping and Reactor efficiency

| | C2_e_recovery |
|---|---|
| count | 120.000000 |
| mean | 0.087574 |
| std | 0.088757 |
| min | 0.000000 |
| 25% | 0.006587 |
| 50% | 0.064132 |
| 75% | 0.123741 |
| max | 0.279625 |



# Data Preprocessing




### Parse data headers

Takes the headers of each column facilitating the creation of feature matrix

### Handle Missing Data

Converts data into numeric format, hence cleaning NaNs, find missing or inconsistent data values. The dataset reduces from 282 to 26 samples.

Complete set $\mathcal{C} = \{i : \forall j \in [1, d], X_{ij} \neq \text{NaN}\}$

Clean data $X_{\text{clean}} = X[\mathcal{C}], \quad y_{\text{clean}} = y[\mathcal{C}]$

$$\tilde{x}_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}$$

### Standardization

Transforms features to have 0 mean and unit variance; SVM model will be symmetric.

Reactor type, y $y \in \{0, 1, 2\}^n$

### Feature Selection

Selects 6 key features from the 14: C2, C2_e_recovery, OD, pH, Ni-C, day

Given n clean samples and d features,

$X \in \mathbb{R}^{n \times d}, \quad X_{ij} = (\text{i-th sample, j-th feature})$

# Train-Test Split

$$\text{Stratified Split} : \min_{\text{split}} \sum_{k=0}^{2} |p_k^{\text{train}} - p_k^{\text{full}}|$$

## Stratified K-fold Cross Validation:

- Data split: 70% train (88), 30% test (38)

## Class Balance:

- Data samples: 126 (equally split for 3 reactors)
- Train - per class ~ 29 – 30 samples
- Test - per class ~ 12 - 13 samples

# SVM Model

## What is SVM?!

- Supervised classification learning algorithm
- Finds hyperplane that maximizes margin between classes.

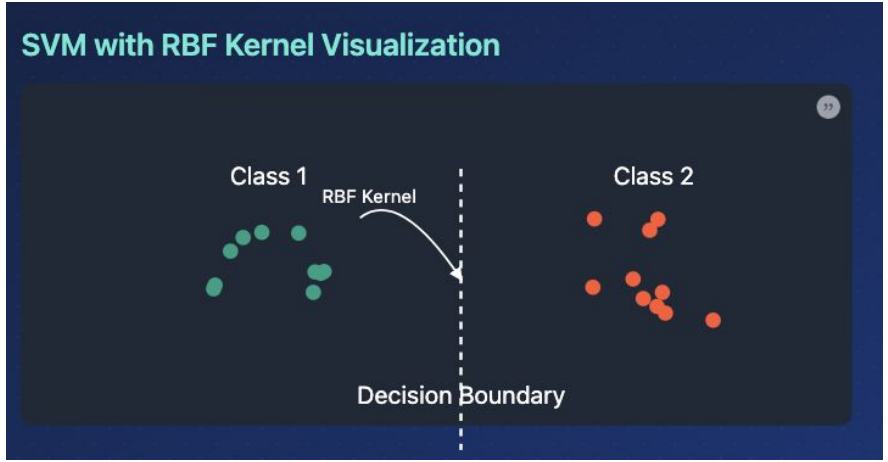
## RBF Kernel?!

$$K(x_i, x_j) = \exp\left(-\gamma \|x_i - x_j\|_2^2\right)$$

- Handles non-linear decision boundaries
- Maps data to higher dimension

## Key Parameters?!

- **C** - regularization parameter
- $\gamma$ - kernel bandwidth




Given labeled data $(x_i, y_i)_{i=1}^n$, SVM solves the optimization problem:

$$\min_{\mathbf{w},b,\boldsymbol{\xi}} \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{n} \xi_i$$

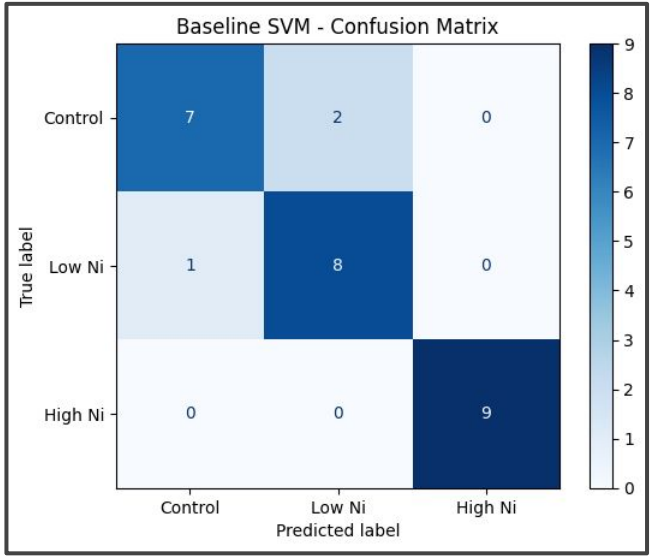
subject to:

$$y_i(\mathbf{w}^\top \phi(x_i) + b) \geq 1 - \xi_i, \quad \xi_i \geq 0 \quad \forall i$$

# Evaluation Metrics (Baseline)

## Confusion Matrix:

$$C \in \mathbb{Z}^{3\times 3}, \quad C_{ij} = \#\{n : y_n = i, \hat{y}_n = j\}$$




**Training accuracy:** 0.8730

**Test accuracy:** 0.8889

Classification Report (Test Set):

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| Control | 0.88 | 0.78 | 0.82 | 9 |
| Low Ni | 0.80 | 0.89 | 0.84 | 9 |
| High Ni | 1.00 | 1.00 | 1.00 | 9 |
| accuracy | | | 0.89 | 27 |
| macro avg | 0.89 | 0.89 | 0.89 | 27 |
| weighted avg | 0.89 | 0.89 | 0.89 | 27 |

# Hyperparameter Optimization

## GridSearchCV:

$$\boxed{\mathcal{G} = \{0.1, 1, 10, 100\} \times \{0.001, 0.01, 0.1, \text{'scale'}, \text{'auto'}\}}$$

- It evaluates the following, to determine the best pair of hyperparameters.

## CV-Score:

- For each (C, $\gamma$) pair, estimate generalization error via cross-validation:

$$\boxed{\text{CV-Score} = \frac{1}{k} \sum_{i=1}^{k} F_1(M_{-i}, X_{\text{val}}^{(i)}, y_{\text{val}}^{(i)})}$$

# Evaluation Metrics (Optimized)

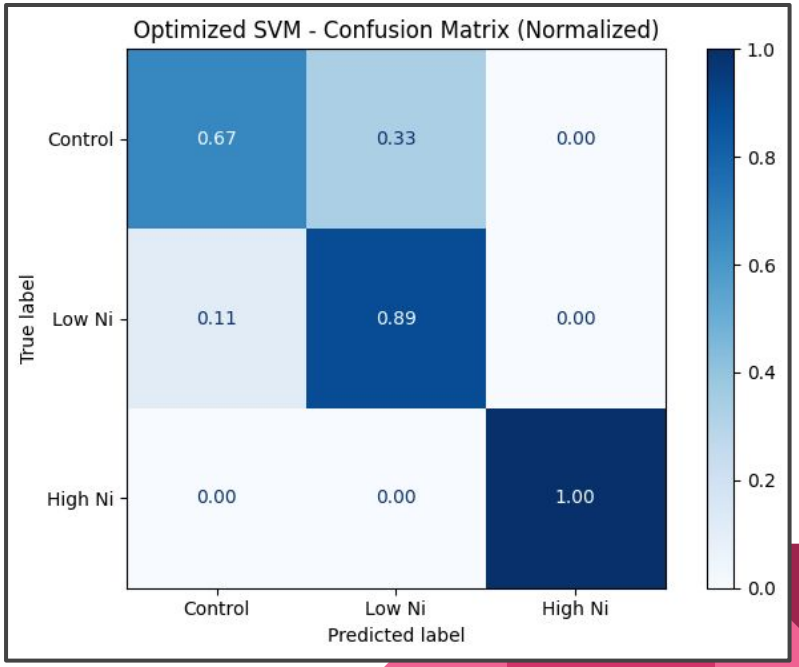
**GridSearch Results:**

- Best parameters: {C: 10; $\gamma$: 0.1; kernel: rbf}
- Best CV F1-score: 0.7857

$$\boxed{\mathrm{F1} = 2 \times \frac{\mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}}}$$

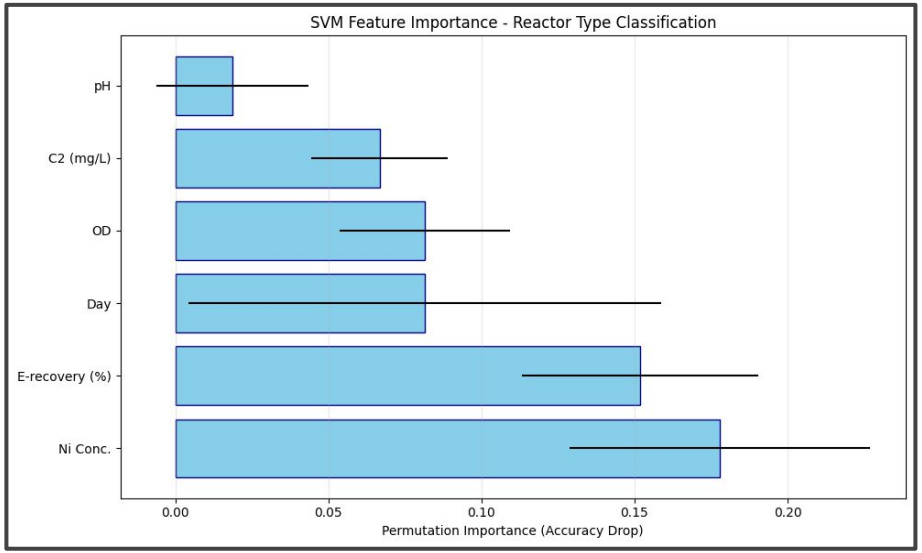
**Optimized SVM:**

- Training accuracy: 0.8889
- Test accuracy: 0.8519
- Test F1-score (weighted): 0.8500



# Model Insights

- The weightage of each feature in the classification.




**Final results**

```
Support Vector Analysis:
Number of support vectors: [16 18  6]
Support vectors per class:
  Control: 16 support vectors
  Low Ni: 18 support vectors
  High Ni: 6 support vectors

Misclassification Analysis:
Total misclassified: 4 out of 27
Error rate: 14.81%
```

# THANK YOU!